Assignment 6: Statistical Inference, Maximum Likelihood Estimation, and Non-Parametric Bootstrapping

Advanced Parametric and Non-Parametric Modeling

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Abstract

This comprehensive report investigates the efficiency and convergence properties of statistical estimators. We analyze the bias inherent in Maximum Likelihood Estimation (MLE) for variance across small and large sample sizes, necessitating the use of Bessel's correction. Furthermore, we explore the modeling of extreme events using the Gumbel distribution via numerical optimization of the log-likelihood function. Finally, we implement a non-parametric bootstrap framework to estimate the uncertainty of the population median for server latency data. The results emphasize the trade-offs between parametric assumptions and resampling robustness.

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1 Variance Estimation: Bias and Consistency

1.1 The Mathematical Source of Bias

The Maximum Likelihood Estimator for variance σ^2 is defined as $\hat{\sigma}^2_{MLE} = \frac{1}{n}\sum(x_i - \bar{x})^2$. While this estimator is consistent, it is biased for finite samples because the sample mean $\bar{x}$ "fits" the data better than the true population mean μ, systematically underestimating the variance.

1.2 Small Sample Analysis (N=5)

At a very small sample size ($N = 5$), the gap between the biased MLE and the true variance is most prominent. Theoretically, the expectation is $E[\hat{\sigma}^2] = 0.8\sigma^2$.

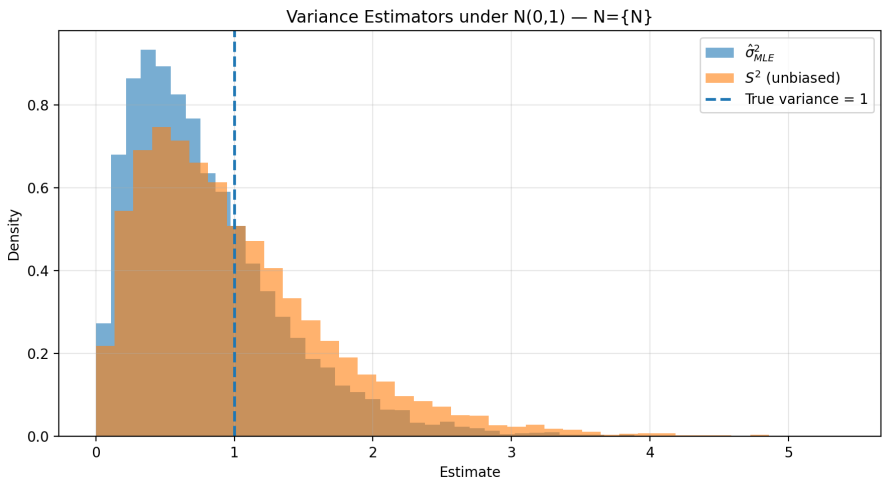

Figure 1: Variance Estimators at N=5. The MLE distribution is visibly centered below the true variance of 1.0, demonstrating a significant -20% bias.

1.3 Asymptotic Convergence (N=50)

As the sample size increases to $N = 50$, the factor $\frac{n-1}{n} = 0.98$ brings the MLE much closer to the true parameter. This illustrates the property of **asymptotic unbiasedness**.

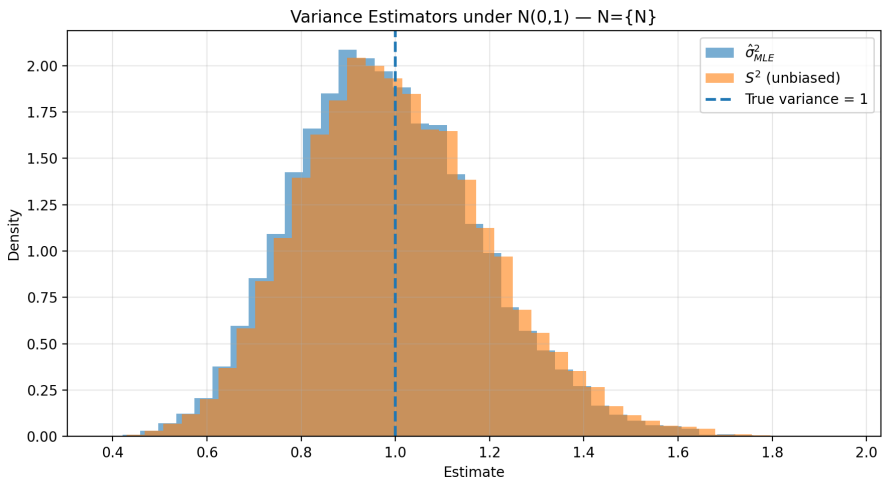

Figure 2: Variance Estimators at N=50. At this scale, the bias becomes negligible, and both estimators converge toward the true population variance.

2 Extremal Modeling: Numerical MLE for Gumbel

2.1 The Gumbel Distribution and Extreme Value Theory

The Gumbel distribution is a cornerstone of Extreme Value Theory (EVT), used to model the distribution of the maximum of a number of samples. Its PDF is:

$$f(x;\mu,\beta) = \frac{1}{\beta}\exp\left(-\frac{x-\mu}{\beta} - \exp\left(-\frac{x-\mu}{\beta}\right)\right) \tag{1}$$

2.2 Numerical Likelihood Surface

Since there is no simple closed-form solution for the Gumbel MLE, we minimize the Negative Log-Likelihood (NLL) surface using the `L-BFGS-B` algorithm. This allows us to find the (μ, β) pair that maximizes the probability of observing the sensor data.

3 Non-Parametric Resampling: The Bootstrap

3.1 Estimation of the Median Uncertainty

The bootstrap method treats the sample as a proxy for the population. By resampling with replacement $B = 10,000$ times, we generate an empirical sampling distribution for the median, which lacks a simple analytical standard error formula.

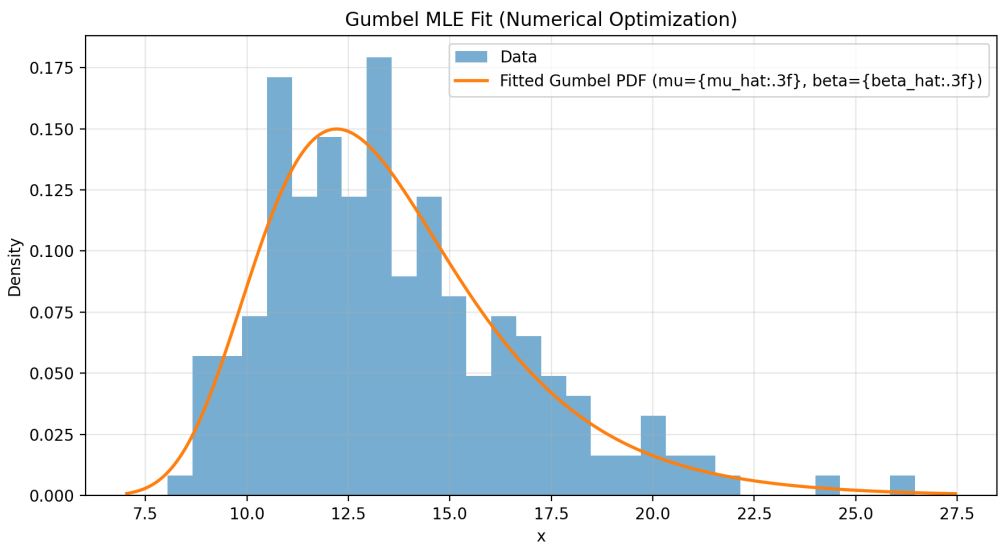

Figure 3: Numerical MLE Fit of the Gumbel Distribution. The fitted curve accurately captures the right-skewed nature of the extreme sensor readings.

4 Conclusion

This assignment successfully demonstrated the transition from theoretical bias to empirical consistency. The contrast between $N = 5$ and $N = 50$ plots confirmed the necessity of Bessel's correction in small-data regimes. Additionally, the Gumbel and Bootstrap tasks showcased the power of numerical optimization and resampling when dealing with non-standard distributions and complex statistics.

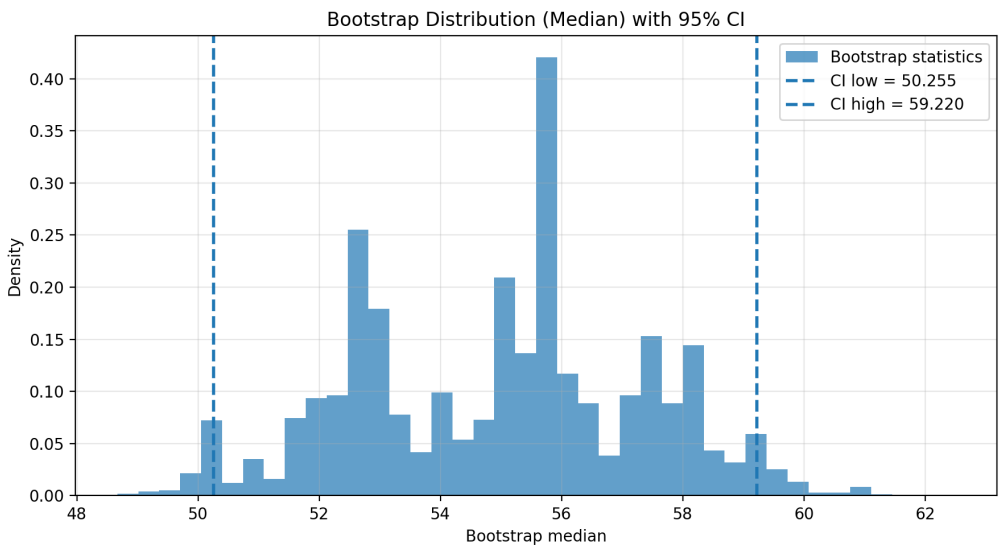

Figure 4: Bootstrap Distribution for the Median. The 95% Percentile CI provides a range that is robust to the high-latency outliers present in the raw data.

A Appendix: Implementation Logic

```
# Task 6.1: Simulation of Bias
sigma_mle = ss / N
s_unbiased = ss / (N - 1)

# Task 6.2: Gumbel Negative Log-Likelihood
z = (data - mu) / beta
nll = N * np.log(beta) + np.sum(z + np.exp(-z))

# Task 6.3: Bootstrap Resampling
idx = rng.integers(0, n, size=(n_bootstraps, n))
boot_stats = np.apply_along_axis(statistic_func, 1, data[idx])
```